LABSHEET-01

1.Print your current working directory to verify your location in the filesystem.

pwd

2.List all files and directories in your current location using the basic ls command.

ls

3.List all contents of your home directory, including hidden files (starting with a dot), using appropriate flags.

ls -la ~

4.Display all files in long-listing format with human-readable file sizes using ls.

ls -lh

5. Create a new directory named myfirstlab in your current directory.

mkdir myfirstlab

6. Navigate into the newly created myfirstlab directory using cd.

cd myfirstlab

7. Inside myfirstlab, create three empty text files: a.txt, b.txt, and c.txt using a single command.

touch a.txt b.txt c.txt

8. Display the full contents of the system file /etc/os-release on your terminal screen.

cat /etc/os-release

9. Check the username of the user currently logged into the terminal session.

whoami

10. Display your system's hostname on the terminal.

hostname

11. Check how much disk space is currently free on your system in human-readable format.

df -h

12. Check the total system RAM and current memory usage in human-readable units.

free -h

13. Check how long your Linux system has been continuously running.

uptime

14. Display full system and kernel operating information using uname.

uname -a

15. Use the manual pages to look up the documentation for the ls command.

man ls

16. Display a short quick-usage summary for the cp command using the --help flag.

cp --help

17. Use the what is command to get a one-line description of the mv command.

whatis mv

18. Navigate back to your home directory using the shorthand tilde (~) symbol.

cd ~

19. Create a nested folder hierarchy project/src/data in a single command using.

mkdir -p project/src/data

20. Delete the file c.txt you created inside the myfirstlab directory.

rm myfirstlab/c.txt

21. Create a file named students.txt and add 5 student names (one per line) using echo and the append redirection operator (>>).

echo "Aman" > students.txt

echo "Riya" >> students.txt

echo "Rahul" >> students.txt

echo "Priya" >> students.txt

echo "Neha" >> students.txt

22. Display only the first 2 names from students.txt using the head command.

head -n 2 students.txt

23. Display only the last 2 names from students.txt using the tail command.

tail -n 2 students.txt

24. Create a directory named backup and copy students.txt into it without naviga□ng out of your currentdirectory.

mkdir -p backup

cp students.txt backup/

25. Rename students.txt to class_list.txt using the mv command.

mv students.txt class_list.txt

26. Use grep to search inside class_list.txt for any name containing a specific letier (e.g., 'a' or 'R').

grep 'a' class_list.txt

grep -i 'a' class_list.txt

27. Display line numbers along with matching text when searching for a string in class_list.txt using grep -n.

grep -n 'a' class_list.txt

28. Change permissions of class_list.txt so only the owner has read and write rights (no execute), and

verify using long listing.

chmod 600 class_list.txt

ls -l class_list.txt

29. Find all files ending with .txt in your home directory tree using the find command.

find ~ -type f -name "*.txt"

30. Use wc -l to count the exact number of lines (students) in class_list.txt.

wc -l class_list.txt

31. Display the first 5 lines of /etc/passwd on the terminal.

head -n 5 /etc/passwd

32. Display the last 5 lines of /var/log/dpkg.log or /var/log/syslog.

tail -n 5 /var/log/dpkg.log

tail -n 5 /var/log/syslog

33. Copy an en□re directory structure (project/) into a new folder named project_backup/ recursively.

cp -r project/ project_backup/

34. View a large text file page-by-page using less and identify the key used to quit the pager.

less /var/log/dpkg.log

35. Move two files (a.txt and b.txt) simultaneously into a target folder named archive/ Hard.

mkdir -p archive

mv a.txt b.txt archive/

36. Combine commands: Create a directory logs, inside it create app.log with 10 lines of dummy text using a loop or repeated echo >>, then view the last 3 lines using tail.

mkdir -p logs

for i in {1..10}; do echo "Log entry $i"; done > logs/app.log

tail -n 3 logs/app.log

37. Use find paired with -exec to locate all .log files in your current working path and display their contents in a single command line.

find . -type f -name "*.log" -exec cat {} \;

38. Recursively search all files in your current directory tree for the word 'error' using grep -r.

grep -riw "error" .

39. Create a shell script file test.sh, grant it execute permissions using symbolic chmod syntax (+x), and write a comment inside explaining the permission string rwxr-xr--.

touch test.sh

echo '#!/bin/bash' > test.sh

echo '# rwxr-xr-- means owner=rwx, group=r-x, others=r--' >> test.sh

chmod +x test.sh

ls -l test.sh

chmod 754 test.sh

40. Write a single-line command using chaining (&&) to create a folder demo, enter it, and list its contents.

mkdir demo && cd demo && ls

41. Combine find and wc -l in a single pipeline to count how many total .txt files exist under your home

directory tree.

find ~ -type f -name "*.txt" | wc -l

42. Find all directories (not regular files) inside /var/log using find -type d.

find /var/log -type d

find /var/log -mindepth 1 -type d

43. Search for a specific file system configuration entry inside /etc/fstab using grep, ignoring case

sensitivity.

grep -i "UUID" /etc/fstab

44. Check the disk space consumption of a specific folder in human-readable format using du -sh.

du -sh /var/log

45. Write a command that displays the 3 largest files in a directory by sor□ng ls output by file size.

ls -lhS /path/to/directory | head -n 4

find /path/to/directory -maxdepth 1 -type f -printf '%s %p\n' | sort -nr | head -n 3

46. Debug / Safety Habit: Explain why running rm -rf * in an unverified path is hazardous, and state the

safety verification command habit that prevents accidental deletion.

rm -rf *

pwd

ls -la

47. Use touch on an existing file to update its access and modificaton timestamp to the current systemtime without modifying its contents.

touch a.txt

48. Create a file containing multiple duplicate lines, then use sort and uniq to display only unique lines from the file.

printf "apple\nbanana\napple\norange\nbanana\n" > fruits.txt

sort fruits.txt | uniq

49. Redirect the stdout of a command (e.g., uname -a) to a file named sys_info.txt, then view its content.

uname -a > sys_info.txt

cat sys_info.txt

50. Find all files in /var/log that were modified within the last 7 days using find -mtime

find /var/log -type f -mtime -7